

Homework #1

CS5402 — Intro To Data Mining

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Due Date: July 6th, 2018

1 1-Rule Method

Problem Statement. Use the 1-rule (1R) method to find the best single attribute to determine restaurant. In order to demonstrate that you actually know how this method works (and aren't just guessing at which attribute is best), you must fill in ALL of the blank values in the table below; otherwise, you will not receive any credit for this problem.

Attribute	Attribute Value	# Rows	Most Frequent Value	Errors	Total Errors
Meal Preference	Hamburger	3	McDonalds (3)	0	2
	Fish	2	Burger King (2)	0	
	Chicken	4	Wendy's (2)	2	
Gender	M	5	Burger King/McDonald's (2/2)	3	5
	F	4	McDonald's (2)	2	
Drink Preference	Pepsi	3	Burger King (2)	1	3
	Coke	6	McDonald's (4)	2	

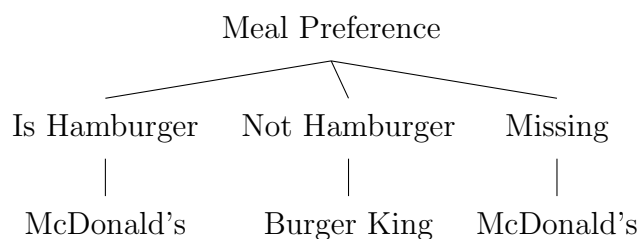
From this, the following rules would be generated:

Meal Preference = *Hamburger* $\implies$ McDonald's
Meal Preference = *Fish* $\implies$ Burger King
Meal Preference = *Chicken* $\implies$ Wendy's

2 Decision Stump

Problem Statement. Create the dataset given in problem 1. as an arff or csv file, and run DecisionStump on it in Weka. List the classification rules that are produced by Weka (you can just include a screenshot of your Weka output) AND hand-draw the decision tree that corresponds to those rules.

From Weka's Decision Stump, we get the following output:
 For the figure, we get the following decision tree:



3 Statistical Modeling

Problem Statement. *Statistical modeling can be used to compute the probability of occurrence of an attribute value. Based on the data given in the table below, if we have a new instance where $ageGroup = youngAdult$, $gender = M$, and $bookPreference = nonFiction$, what is the likelihood that $musicPreference = country$? Just set up the equation to compute this with the appropriate values; you don't have to actually calculate the final answer.*

$$\text{Likelihood of Country} = \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{3}{8} = \frac{1}{36} \approx 0.02\bar{7}$$

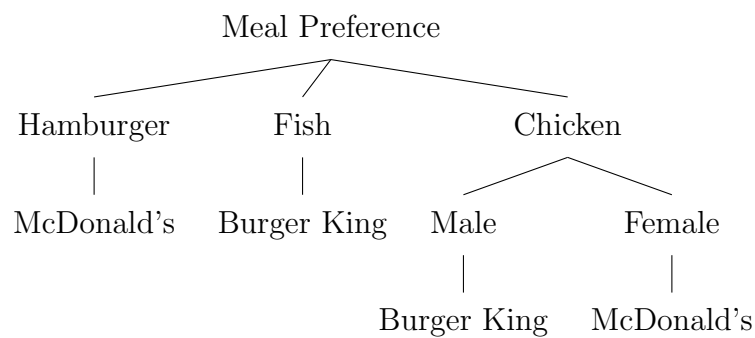
4 Id3

Problem Statement. *Create the dataset given in problem 1. as an arff or csv file, and run Id3 on it in Weka. Include a screenshot of the decision tree output that is produced by Weka AND hand-draw the decision tree. Note: You may have to install the package containing Id3.*

From Weka's Id3, we get the following output:



For the figure, we get the following decision tree:



5 Entropy

Problem Statement. *If we want to make a decision tree for determining restaurant, we must decide which of the three non-decision attributes (mealPreference, gender, or drinkPreference) to use as the root of the tree.*

- *Set up the equation to compute what in lecture we called `entropyBeforeSplit` for restaurant. You do not have to actually solve (i.e., calculate the terms in) the*

equation, just set up the equation with the appropriate values.

$$\text{Entropy Before Split} = -3/10 \log_2 (3/10) - 5/10 \log_2 (5/10) - 2/10 \log_2 (2/10) \approx 1.485$$

- Set up the equation to compute entropy for mealPreference when its value is chicken. That is, a tree with mealPreference at the root would have three branches (one for hamburger, one for chicken, and one for fish), requiring us to compute entropyHamburger, entropyChicken, and entropyFish; here we only want you to set up the equation to compute entropyChicken. You do not have to actually solve (i.e., calculate the terms in) the equation, just set it up using the appropriate values.

$$\text{Entropy Chicken} = -1/4 \log_2 (1/4) - 1/4 \log_2 (1/4) - 2/4 \log_2 (2/4) \approx 1.040$$

- Suppose that instead of considering mealPreference to be the root of this decision tree, we had instead considered drinkPreference. Set up the equation to compute information gain for drinkPreference given the variables specified below.

$$\text{Information Gain} = X - (7/10 C + 3/10 P)$$

6 isSticky

Problem Statement. Create the dataset shown below as an arff or csv file and run the Prism algorithm on it in Weka specifying isSticky as the decision attribute. List the classification rules that are produced (you can just include a screenshot of your Weka output). Then work out the Prism algorithm by hand to show what classification rules you would get; who knows, they might be different!

6.1 Is Sticky = No

consistency	packaging	chocolate	roundShaped	isSticky
soft	individuallyWrapped	yes	no	no
soft	box	yes	no	no
hard	sack	no	yes	no
hard	box	yes	yes	no
hard	individuallyWrapped	no	no	yes
hard	box	no	yes	yes
soft	individuallyWrapped	no	no	yes

Which leads us to the following T and P values:

6.2 Is Sticky = No

Label	T	P	T/P
soft	3	2	0.66
hard	4	2	0.50
individually wrapped	3	1	0.33
box	3	2	0.66
sack	1	1	1
chocolate	3	3	1
not chocolate	4	1	0.25
round	3	2	0.66
not round	4	2	0.5

This leads to the following rule:

$$\text{chocolate} = \text{yes} \implies \text{is sticky} = \text{no}$$

Reducing Our dataset to:

6.3 Is Sticky = No

consistency	packaging	chocolate	roundShaped	isSticky
hard	sack	no	yes	no
hard	individuallyWrapped	no	no	yes
hard	box	no	yes	yes
soft	individuallyWrapped	no	no	yes

Which leads us to the following T and P values:

6.4 Is Sticky = No

consistency	packaging	chocolate	roundShaped	isSticky
hard	sack	no	yes	no
hard	individuallyWrapped	no	no	yes
hard	box	no	yes	yes
soft	individuallyWrapped	no	no	yes

Which leads us to the following T and P values:

Label	T	P	T/P
soft	1	0	0
hard	3	1	0.33
individually wrapped	2	0	0
box	1	0	0
sack	1	1	1
chocolate	0	0	
not chocolate	4	1	0.25
round	2	1	0.5
not round	2	0	0

Which leads us to the following rule:

$$\text{packaging} = \text{sack} \implies \text{is sticky} = \text{no}$$

6.5 Is Sticky = Yes

hard	individuallyWrapped	no	no	yes
hard	box	no	yes	yes
soft	individuallyWrapped	no	no	yes

Which leads us to the following T and P values:

6.6 Is Sticky = No

Label	T	P	T/P
soft	1	1	1
hard	2	2	1
individually wrapped	2	2	1
box	1	1	1
sack	0	0	–
chocolate	0	0	–
not chocolate	3	3	1
round	1	1	1
not round	2	2	1

Which leads us to the following rule:

$$\text{chocolate} = \text{no} \implies \text{is sticky} = \text{yes}$$

6.7 Final Ruleset

From this, the following rules would be generated:

chocolate = yes $\implies$ is sticky = no
packaging = sack $\implies$ is sticky = no
chocolate = no $\implies$ is sticky = yes

6.8 Weka Output

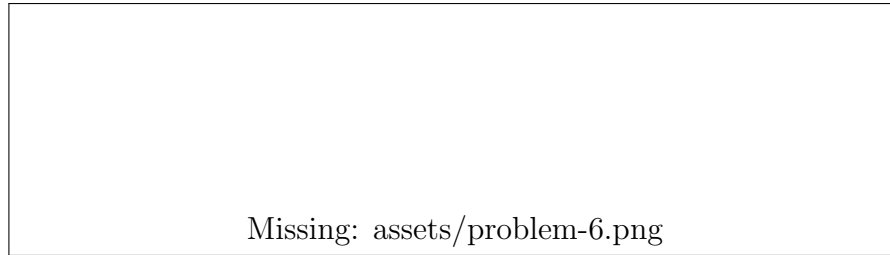


Figure 1: Weka output for problem 6